

ATOMIC ENERGY CENTRAL SCHOOL NO.4
RAWATBHATA

CONFIDENCE EXAMINATION-I DECEMBER 2022

Class 12 - Physics

Time Allowed: 3 hours

Maximum Marks: 70

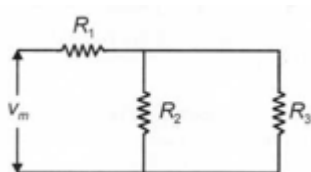
General Instructions:

1. There are 35 questions in all. All questions are compulsory.
2. This question paper has five sections: Section A, Section B, Section C, Section D and Section E. All the sections are compulsory.
3. Section A contains eighteen MCQ of 1 mark each, Section B contains seven questions of two marks each, Section C contains five questions of three marks each, section D contains three long questions of five marks each and Section E contains two case study based questions of 4 marks each.
4. There is no overall choice. However, an internal choice has been provided in section B, C, D and E. You have to attempt only one of the choices in such questions.
5. Use of calculators is not allowed.

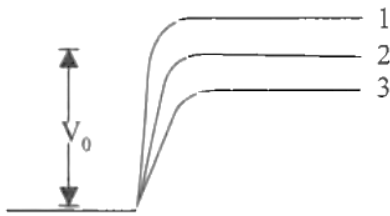
Section A

1. The barrier potential of a p-n junction diode does not depend on [1]
 - a) temperature
 - b) forward bias
 - c) doping density
 - d) diode design

2. For ensuring dissipation of same energy in all three resistors (R_1 , R_2 , R_3) connected as shown in figure, their values must be related as [1]



- a) $R_2 = R_3$ and $R_1 = \frac{R_2}{4}$
 - b) $R_1 = R_2 = R_3$
 - c) $R_2 = R_3$ and $R_1 = 4R_2$
 - d) $R_1 = R_2 + R_3$
3. A concave and a convex lens have the same focal length of 20 cm and are put into contact to form a lens combination. The combination is used to view an object of 5 cm length kept at 20 cm from the lens combination. As compared to the object, the image will be: [1]
 - a) magnified and inverted
 - b) of the same size as the object and would be erect
 - c) of the same size as the object but would be inverted
 - d) reduced and erect
 4. In the figure, V_0 is the potential barrier across a p-n junction, when no battery is connected across the junction. [1]



- a) 1 and 3 both correspond to forward bias of junction.
- b) 3 and 1 both correspond to reverse bias of junction.
- c) 1 corresponds to forward bias and three corresponds to reverse bias of junction.
- d) 3 corresponds to forward bias of junction and one corresponds to reverse bias of junction.

5. The parallel combination of two air-filled parallel plate capacitors of capacitance C and nC is connected to a battery of voltage, V . When the capacitors are fully charged, the battery is removed and after that a dielectric material of dielectric constant K is placed between the two plates of the first capacitor. The new potential difference of the combined system is: [1]

- a) $\frac{V}{K+n}$
- b) V
- c) $\frac{(n+1)V}{(K+n)}$
- d) $\frac{nV}{K+n}$

6. The magnetic moment of a current (I) carrying circular coil of radius (r) and number of turns (n) varies as [1]

- a) $\frac{1}{r}$
- b) r^2
- c) $\frac{1}{r^2}$
- d) r

7. Current in a circuit falls from 5 A to 0 A in 0.1 s. If an average emf of 200 V is induced, the self-inductance of the circuit is [1]

- a) 5H
- b) 4H
- c) 2H
- d) 3H

8. The energy of the ground electronic state of hydrogen atom is -13.6 eV. The energy of the first excited state will be [1]

- a) -3.4 eV
- b) -52.4 eV
- c) -6.8 eV
- d) -27.2 eV

9. Monochromatic light of wavelength 589 nm is incident from air on a water surface. What is the speed of refracted light? Refractive index of water is 1.33. [1]

- a) 4.26×10^8 m/s
- b) 5.26×10^8 m/s
- c) 3.26×10^8 m/s
- d) 2.26×10^8 m/s

10. Match Column I with Column II with appropriate matching. [1]

Column I	Column II
a. λ	i. L^2MT^{-2}
b. ρ	ii. electric field intensity
c. Torque	iii. $\frac{q}{V}$
d. electric field	iv. $\frac{q}{L}$

-
- a) (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii) b) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)
- c) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii) d) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)
11. Resistance of the semiconductor: [1]
- a) decreases with temperature b) increases with temperature
- c) remains unaffected with temperature d) None of these
12. Critical angle of light passing from glass to air is minimum for: [1]
- a) Red b) Green
- c) Yellow d) Violet
13. An electron accelerated under a potential difference V volt has a certain wavelength λ . Mass of proton is some 2000 times of the mass of the electron. If the proton has to have the same wavelength λ , then it will have to be accelerated under a potential difference of: [1]
- a) $\frac{V}{2000}$ volt b) V volt
- c) $\sqrt{2000}$ V volt d) 2000 V volt
14. When two charged conductors are connected by a wire, [1]
- a) there is always gain of energy b) there is loss or gain of energy depending upon the potentials
- c) there will not be any change in energy d) there is always loss of energy
15. A parallel monochromatic beam of light is incident normally on a narrow slit. A diffraction pattern is formed on a screen placed perpendicular to the direction of the incident beam. At the first minimum of the diffraction pattern, the phase difference between the rays coming from the two edges of the slit is [1]
- a) 2π b) π
- c) $\frac{\pi}{2}$ d) zero
16. **Assertion (A):** Thermonuclear fusion reactions may become the source of unlimited power for mankind. [1]
Reason (R): A single fusion event involving isotopes of hydrogen produces more energy than energy from nuclear fission of ${}_{93}^{235}\text{U}$.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
17. **Assertion (A):** When cooking in microwave ovens, metal containers are used. [1]
Reason (R): The energy of microwaves can be easily transferred to the food through metal.
- a) Both A and R are true and R is the correct explanation of A b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false d) A is false but R is true.
18. **Assertion:** Copper is a paramagnetic substance. [1]
Reason: Paramagnetic substance when placed in a magnetic field are feebly magnetised in a direction opposite to that of the magnetising field.

- a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
- c) Assertion is correct statement but reason is wrong statement.

- b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
- d) If both assertion and reason are false.

Section B

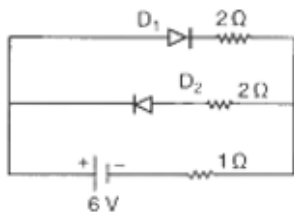
19. Explain the two processes involved in the formation of a p-n junction diode. Hence define the term barrier potential. [2]
20. Calculate the shortest wavelength of the spectral lines emitted in the Balmer series. (Given Rydberg constant, $R = 10^7 \text{ m}^{-1}$) [2]
21. The magnetic field of an electromagnetic wave oscillates parallel to y-axis and is given by $B_y = B_0 \sin(kz - \omega t)$. [2]
- In what direction does the wave travel?
 - Parallel to which axis does the associated electric field oscillate?

OR

Name the electromagnetic radiations used for

- water purification, and
- eye surgery.

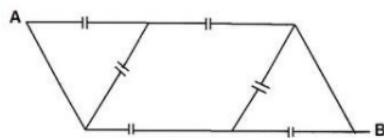
22. For the circuit shown here, find the current flowing through the 1Ω resistor. Assume that the two diodes D_1 and D_2 are ideal diodes. [2]



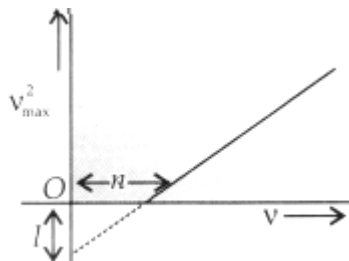
23. Two capacitors of capacitances $3\mu\text{F}$ and $6\mu\text{F}$, are charged to potentials of 2V and 5V respectively. These two charged capacitors are connected in series. Find the potential across each of the two capacitors now. [2]

OR

A network of six identical capacitors, each of value C is made, as shown in Fig. Find the equivalent capacitance between the points A and B is



24. State Einstein's photoelectric equation explaining the symbols used. [2]



Light of frequency ν is incident on a photosensitive surface. A graph of the square of the maximum speed of the electrons ($V_{2\text{max}}$) vs. ν is obtained as shown in the figure. Using Einstein's photoelectric equation, obtain expressions for

- i. Planck's constant,
- ii. Work function of the given photosensitive material in terms of parameters l , n , and mass of the electron m .

25. What is mass defect of a nucleus? Express it mathematically. How do you account for it? [2]

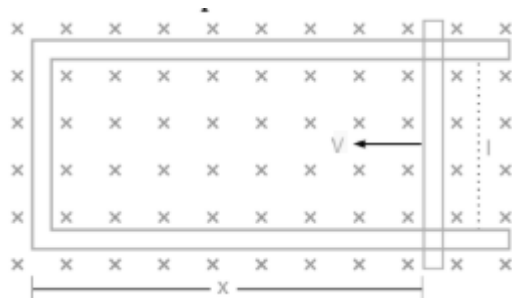
Section C

26. Using Rutherford model of the atom, derive the expression for the total energy of the electron in hydrogen atom. [3]
What is the significance of total negative energy possessed by the electron?

27. What is the effect on the interference fringes in Young's double-slit experiment due to each of the following [3]
operations? Justify your answers.

- i. The screen is moved away from the plane of the slits.
- ii. The separation between slits is increased.
- iii. The source slit is moved closer to the plane of double slit.

28. i. A rod of length l is moved horizontally with a uniform velocity v in a direction perpendicular to its length [3]
through a region in which a uniform magnetic field is acting vertically downward. Derive the expression for the emf induced across the ends of the rod.



ii. How does one understand this motional emf by invoking the Lorentz force acting on the free charge carriers of the conductor? Explain.

OR

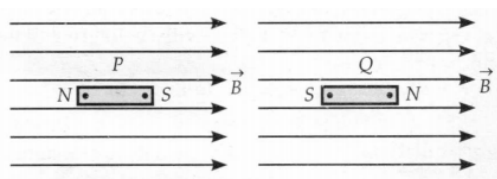
State Lenz's law. Give one example to illustrate this law. "The Lenz's law is a consequence of the principle of conservation of energy." Justify this statement.

29. i. Name the e.m. waves which are used for the treatment of certain forms of cancer. Write their frequency [3]
range.
- ii. Thin ozone layer on top of the stratosphere is crucial for human survival. Why?
- iii. An e.m. wave exerts pressure on the surface on which it is incident. Justify.

OR

Name the type of EM waves having a wavelength range 10^{-7} m to 10^{-9} m. How are these waves generated? Write their two uses.

30. Two identical bar magnets P and Q are placed in two identical uniform magnetic fields as shown in figure. [3]
Justify that both the magnets are in equilibrium. Which one of these is in stable equilibrium? Give reasons for your answer.



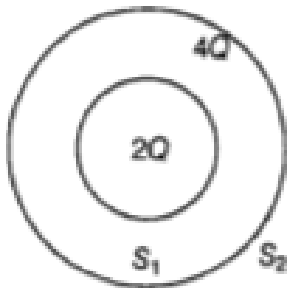
Section D

31. i. Derive an expression for the electric field at any point on the equatorial line of an electric dipole. [5]

- ii. Two identical point charges, q each, are kept $2m$ apart in the air. A third point charge Q of unknown magnitude and sign is placed on the line joining the charges such that the system remains in equilibrium. Find the position and nature of Q .

OR

- a. Deduce the expression for the torque acting on a dipole of dipole moment p in the presence of uniform electric field E .
- b. Consider two hollow concentric spheres S_1 and S_2 enclosing charges $2Q$ and $4Q$ respectively as shown in the figure.
- Find out the ratio of the electric flux through them.
 - How will the electric flux through the sphere S_1 changes, if a medium of dielectric constant ϵ_r is introduced in the space inside S_1 in place of air? Deduce the necessary expression?



32. i. Draw a labelled ray diagram showing the image formation of a distant object by a refracting telescope. [5]
Deduce the expression for its magnifying power when the final image is formed at infinity.
- ii. The sum of focal lengths of the two lenses of a refracting telescope is 105 cm. The focal length of one lens is 20 times that of the other. Determine the total magnification of the telescope when the final image is formed at infinity.

OR

- i. Draw a labelled ray diagram to obtain the real image formed by an astronomical telescope in normal adjustment position. Define its magnifying power.
- ii. You are given three lenses of power 0.5 D, 4D and 10 D to design a telescope.
- Which lenses should be used as objective and eyepiece? Justify your answer.
 - Why is the aperture of the objective preferred to be large?
33. i. Define the term conductivity of a metallic wire. Write its SI unit. [5]
- ii. Using the concept of free electrons in a conductor, derive the expression for the conductivity of wire in terms of number density and relaxation time. Hence obtain the relation between current density and the applied electric field E .

OR

- i. Why do the free electrons, in a metal wire, flowing by themselves, not cause any current flow in the wire? Define drift velocity and obtain an expression for the current flowing in a wire, in terms of the drift velocity of the free electrons.
- ii. Use the above expression to show that the resistivity, of the material of a wire, is inversely proportional to the relaxation time for the free electrons in the metal.

Section E

34. Read the text carefully and answer the questions: [4]

A magnetic field can be produced by moving, charges or electric currents. The basic equation governing the magnetic field due to a current distribution is the Biot-Savart law. Finding the magnetic field resulting from a current distribution involves the vector product, and is inherently a calculus problem when the distance from the current to the field point is continuously changing. According to this law, the magnetic field at a point due to a current element of length $d\vec{l}$ carrying current I , at a distance r from the element is $d\vec{B} = \frac{\mu_0}{4\pi} \frac{I(d\vec{l} \times \vec{r})}{r^3}$. Biot-Savart law has certain similarities as well as differences with Coulomb's law for electrostatic field e.g., there is an angle dependence in Biot-Savart law which is not present in the electrostatic case.

- (i) What is the direction of magnetic field $d\vec{B}$ at a distance r due to a current element $I dl$ when current I passes through a long conductor ?

OR

A long straight wire carries a current along the z-axis. What will be the magnetic field along the Z axis.

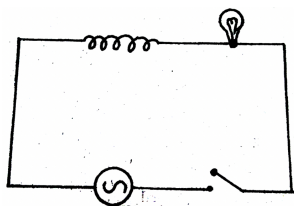
- (ii) What will be the magnetic field due to current carrying wire if observation point lies (a) on the wire (b) perpendicular to wire
- (iii) Two long straight wires are set parallel to each other. Each carries a current i in the same direction and the separation between them is $2r$. What will be the intensity of the magnetic field midway between them?



35. **Read the text carefully and answer the questions:**

[4]

A light bulb and an open coil inductor are connected to an ac source through a key as shown in the figure. The switch is closed and after some time, an iron rod is inserted into the interior of the inductor. The glow of the light bulb (a) increases, (b) decreases, (c) is unchanged, as the iron rod is inserted.



- (i) A bulb and a capacitor are connected in series to an a.c. source of variable frequency. How will the brightness of the bulb change on increasing the frequency of the a.c. source? Give reason.
- (ii) Define capacitor reactance. Write its SI units.
- (iii) A capacitor behaves like a perfect conductor for high frequency a.c. Explain, why?

